

Final Report

Clarify-and-Link: LLM-Enhanced Entity Linking

Semantic Web and Social Networks

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1 Introduction

The project focuses on improving **entity disambiguation** in text, addressing the challenges posed by ambiguous or incomplete mentions. Traditional entity linking methods map mentions directly to knowledge base entities, often failing in ambiguous cases. Our approach introduces a **Clarify-and-Link** method, where a Large Language Model (LLM) generates brief, contextual clarifications for each ambiguous mention, which are appended to the original text. This approach preserves the original context, mitigates hallucination risk, and enables more informed linking decisions.

2 Related Work

Traditional entity linking approaches rely on direct mention-to-entity mapping, often struggling with ambiguity (e.g., “Jordan” could refer to a person, a country, or a river). Recent advancements leverage LLMs for **contextual augmentation**, generating expanded mentions or descriptions. Prior works typically replace mentions with LLM-generated expansions before linking, which can introduce hallucinations. In contrast, our approach appends clarifications, allowing the model to use both the original mention and additional context. This strategy enhances robustness and reliability, inspired by the contextual augmentation paradigm Vollmers et al. [2025].

3 Dataset Description

We conduct experiments using the **AIDA-CoNLL** dataset, a widely used benchmark in entity linking and named entity disambiguation research. AIDA-CoNLL is derived from news articles and provides high-quality annotations of entity mentions linked to a knowledge base (Wikidata). Its combination of realistic, polysemous mentions and diverse context lengths makes it a standard choice for evaluating both baseline and advanced entity linking methods.

Before using this dataset, we performed an in-depth analysis of the data, obtaining the following results:

- **Total mentions:** 23,393 across 946 documents
- **Unique linked entities (QIDs):** 4,081
- **Linked mentions:** 18,541 (79.3%)
- **Average mentions per document:** 24.7

Mention Characteristics:

- Mean length: 9.5 characters; median: 8.0
- Single-word mentions: 14,689 (62.8%). Multi-word mentions: 8,704 (37.2%)
- Ambiguous mentions: 304 (6% of unique mentions), averaging 1.07 entities per mention and maximum ambiguity: 6 entities
- Top ambiguous mentions: “World Cup”, “Foreign Ministry”, “Scotland”
- Entity type distribution: LOC 30.5%, PER 28.0%, ORG 26.9%, MISC 14.6%

Context Characteristics:

- Mean context length: 1,728 characters (328 words). Median: 1,328 characters

Dataset Advantages for Clarify-and-Link: The AIDA-CoNLL dataset is particularly suitable for the Clarify-and-Link approach because it contains a high number of polysemous mentions and short contexts, which are challenging for standard entity linking methods. It also enables direct comparison with Replace-and-Link strategies from original paper and supports evaluation of both disambiguation accuracy and robustness against potential hallucinations introduced by LLM-generated clarifications.

3.1 Dataset Statistics and Visualizations

Table 1: AIDA-CoNLL Dataset Overview

Property	Value
Total mentions	23,393
Unique documents	946
Unique linked entities (QID)	4,081
Linked mentions	18,541 (79.3%)
Average mentions per document	24.7
Mean mention length	9.5 characters
Median mention length	8.0 characters
Multi-word mentions	8,704 (37.2%)
Single-word mentions	14,689 (62.8%)
Ambiguous mentions	304 (6.0%)
Avg entities per mention	1.07
Max ambiguity	6 entities

These analyses highlight the dataset’s distributional properties, the prevalence of ambiguity, and contextual characteristics, as illustrated in Figures 1–8.

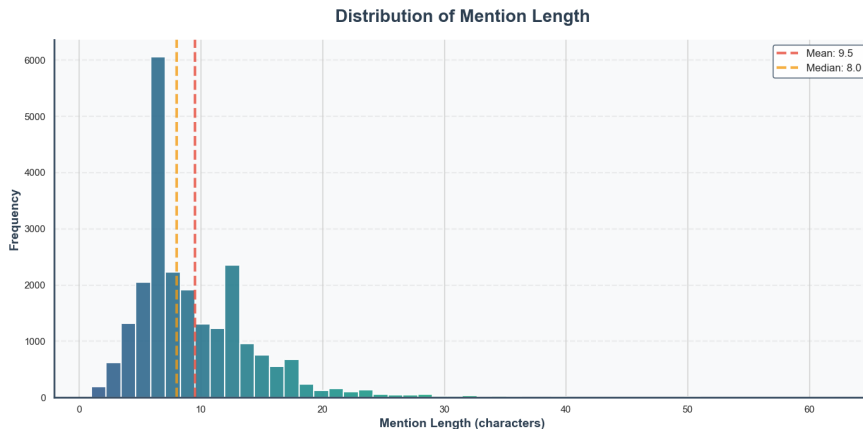


Figure 1: Mention Length Distribution.

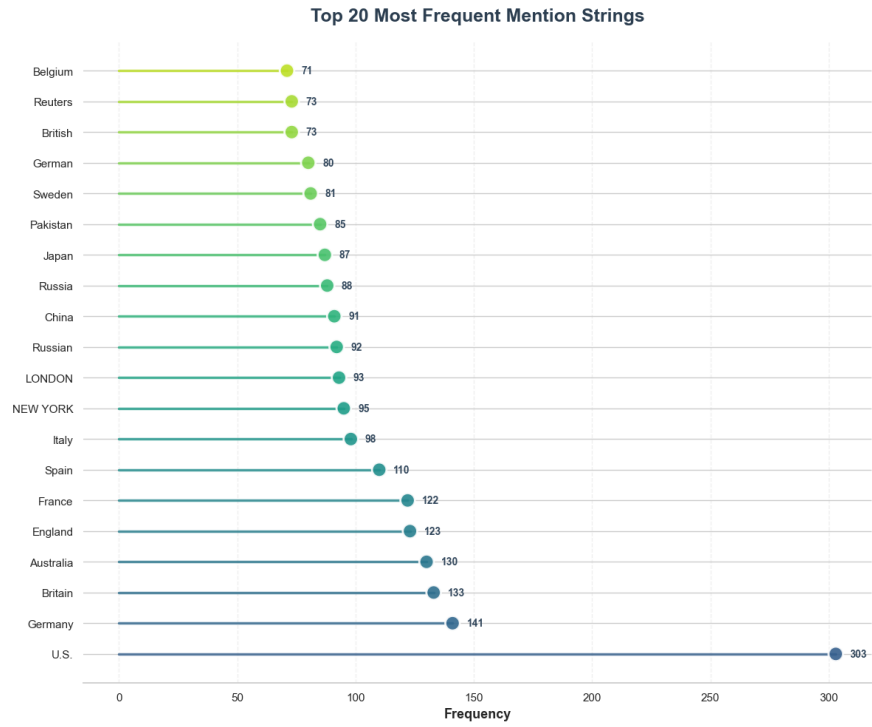


Figure 2: Mention Frequency.

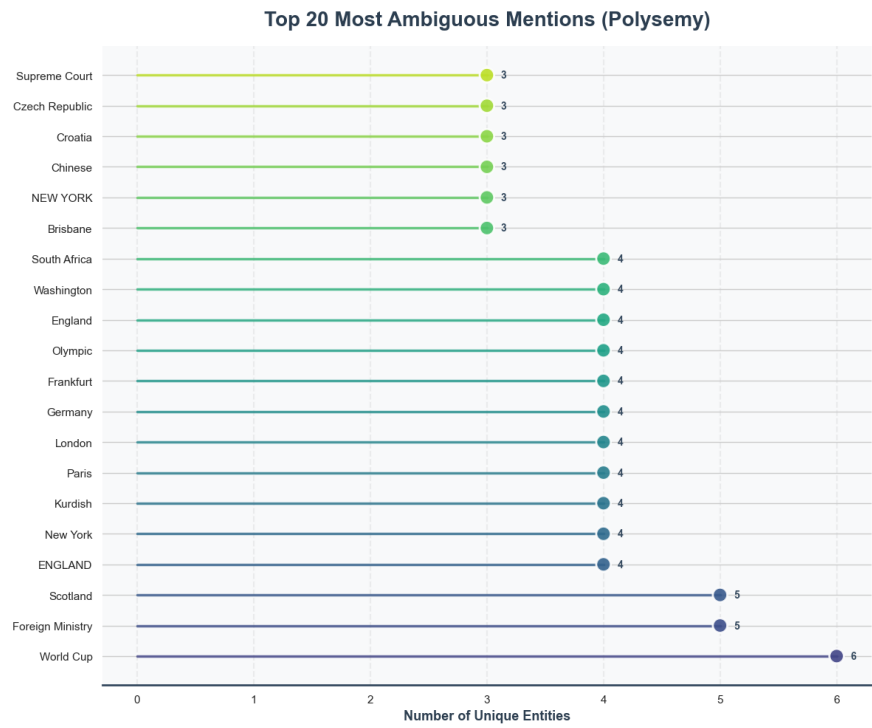


Figure 3: Polysemy Analysis.

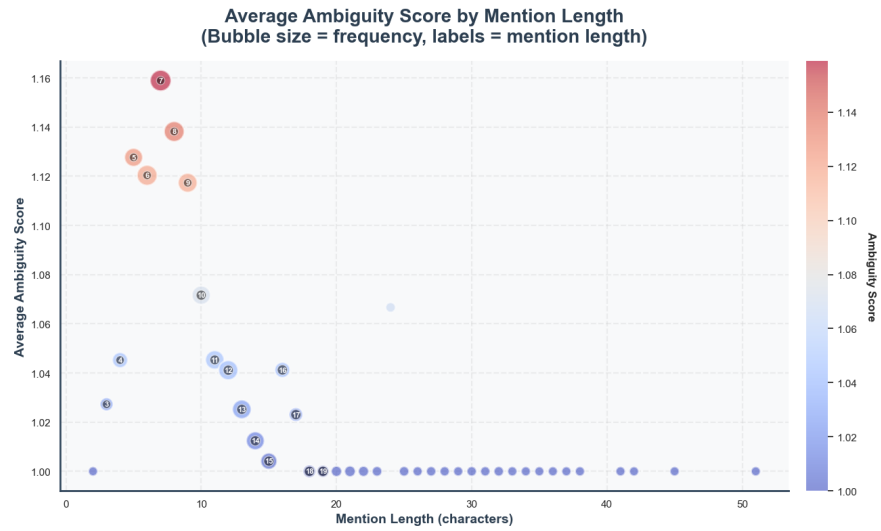


Figure 4: Average Ambiguity by Mention Length.

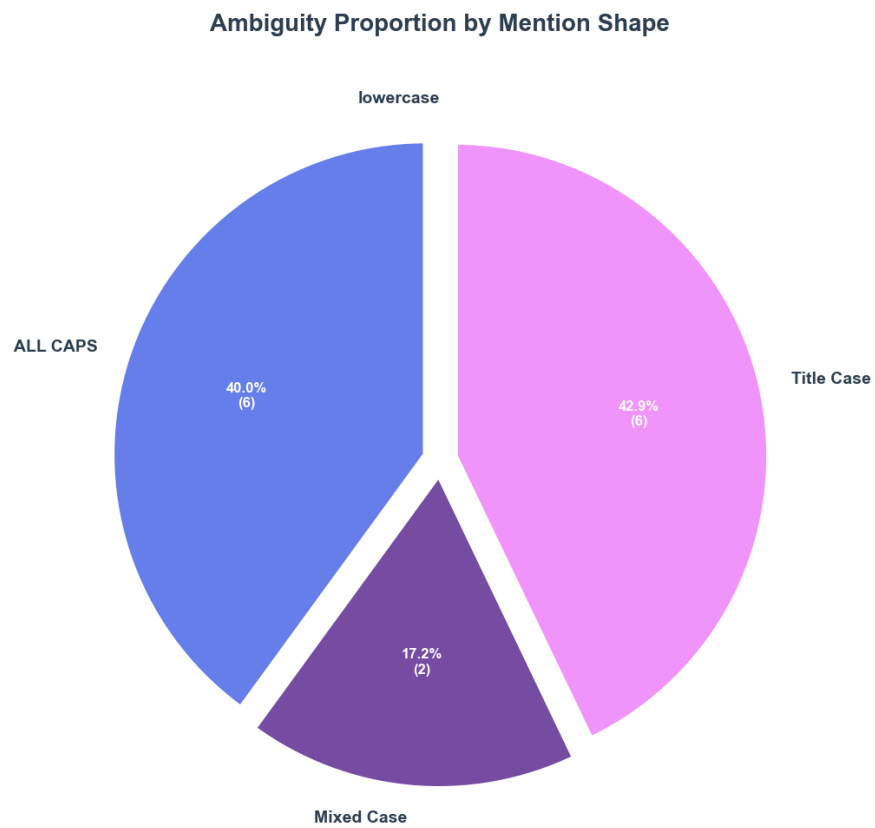


Figure 5: Source of Ambiguous Mentions by Shape.

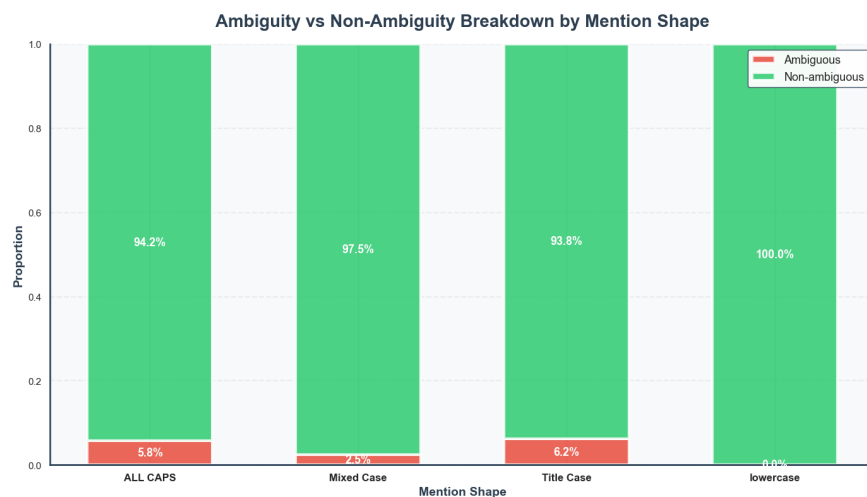


Figure 6: Ambiguity vs Non-Ambiguity Breakdown by Mention Shape.

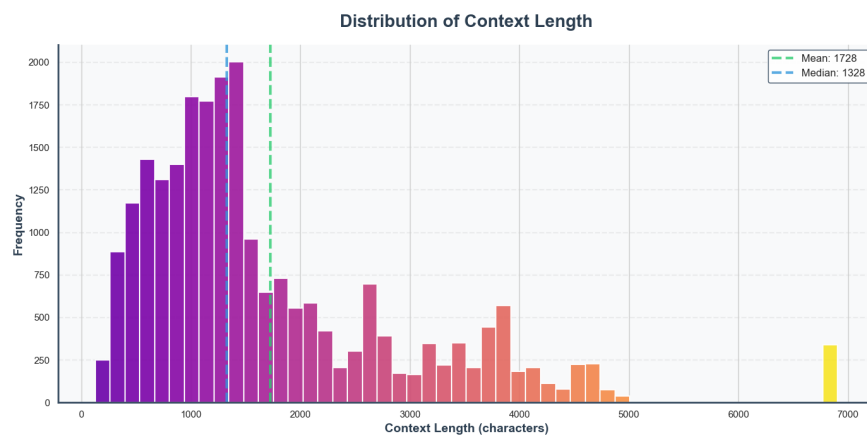


Figure 7: Context Length Distribution.

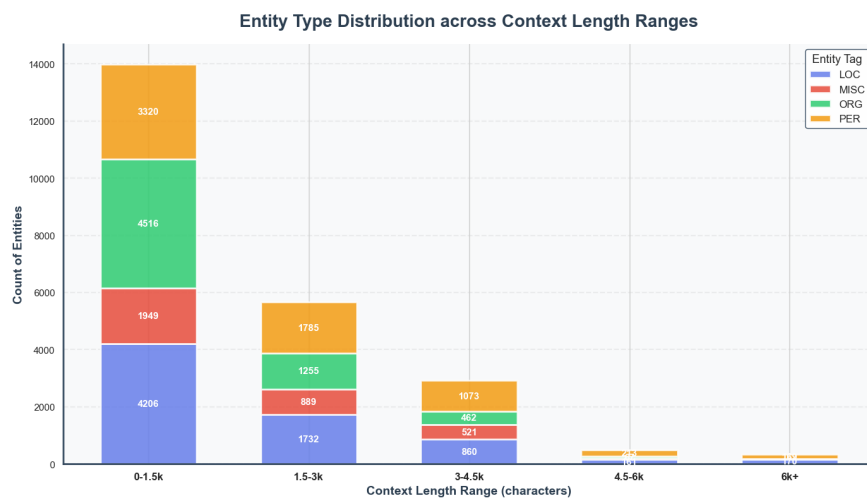


Figure 8: Mention Length vs. Context Length.

4 Method Description

The **Clarify-and-Link** framework enhances entity linking by explicitly introducing an intermediate clarification step generated by a Large Language Model (LLM). The core idea is to augment ambiguous entity mentions with short, factual descriptions derived from their local context, thereby improving downstream disambiguation. The pipeline consists of three stages: data preprocessing, clarification generation, and entity linking model training.

4.1 Stage 1: Data Preprocessing

The AIDA-CoNLL dataset is loaded and split into training, validation, and test partitions. Each document contains raw text together with annotated entity mentions, including character offsets and gold knowledge base identifiers.

A unified preprocessing pipeline is applied to ensure data consistency and usability across all stages:

1. **Context Augmentation:** Each entity mention is enriched with a local context window consisting of surrounding text before and after the mention.
2. **Mention Normalization:** Entity mentions are standardized by lowercasing, trimming whitespace, and removing leading/trailing punctuation.
3. **Conflict Resolution:** Overlapping entity annotations are resolved by retaining only the longest span in each overlapping region.
4. **Chunking Long Documents:** Long documents are optionally split into overlapping text chunks. Entity offsets are adjusted accordingly.

4.2 Stage 2: Clarification Generation

For each unique normalized entity mention, a fixed-size context window (default: 100 characters on each side) is extracted. This context is used to prompt the instruction-tuned LLM **Llama-3.2-3B-Instruct** to generate a concise, factual description of the entity. The model is configured with a maximum output length of 50 tokens and a temperature of 0.3.

4.2.1 Prompt Template

The model is prompted using a structured template. For each entity mention, the left and right context segments are inserted, and the model is instructed to provide factual information for entity linking. The prompt is:

```
prompt = (
    f"<|begin_of_text|><|start_header_id|>system<|end_header_id|>\n\n"
    f"You are creating text augmentations for entity linking. You will provide explanation for\n\n"
    f"highlighted entity in a given context. Provide only useful information. If you can't\n\n"
    f"deduce exactly what the entity denotes, limit yourself to what you can provide. For\n\n"
    f"example, if you recognize the entity denotes a person, provide this information. Do not\n\n"
    f"formulate your answer as a chat message, you will not get any additional information.\n\n"
    f"Provide useful information for entity linking. Your goal is to provide useful information\n\n"
    f"for entity identification and disambiguation.<|eot_id|>"
    f"<|start_header_id|>user<|end_header_id|>\n\n"
    f"Context: {ctx_l} **{mention}** {ctx_r}\n\n"
    f"Describe the entity '{mention}' based on this context.<|eot_id|>"
    f"<|start_header_id|>assistant<|end_header_id|><|end_of_text|>\n\n"
)
```

4.2.2 Efficiency and Robustness Strategies

- **Deduplication:** Queries are made only for unique normalized mentions, with the first observed context reused for subsequent occurrences.
- **Batching:** Clarifications are generated in batches of 32 mentions.
- **Checkpointing:** Intermediate outputs are periodically saved to disk to allow recovery from interruptions.

Generated clarifications are mapped back to original documents. Each document stores the raw text, entity annotations, and a lookup table mapping mentions to clarifications.

4.2.3 Example

Context: "...in the capital of Jordan, visited..."

Mention: "Jordan"

Clarification: "Jordan is a Middle Eastern country located between Israel, Syria, Iraq, and Saudi Arabia."

4.2.4 Implementation Details

- Context windows are truncated symmetrically to focus on relevant information.
- Empty or failed generations are replaced with a fallback string: **Entity:** [mention].
- GPU memory is cleared after each dataset split to prevent out-of-memory errors.

4.3 Stage 3: Entity Linking Model Training

Clarification-augmented documents are converted into sequence-to-sequence training examples for T5-based entity linking models. Two models are trained:

- **Baseline Model:** Uses raw context with entity markers only.
- **Clarified Model:** Uses context augmented with LLM-generated clarifications.

4.3.1 Training Data Construction

4.3.2 Input Format Comparison

Each entity mention is transformed into a single input–output pair:

- **Input:** Context containing entity markers, optionally followed by clarification text.
- **Target:** Gold Wikidata identifier (e.g., Q170566).

Example:

Baseline Input:

link entity: ...in the capital of [START_ENT]Jordan[END_ENT], visited...

Clarified Input:

link entity: ...in the capital of [START_ENT]Jordan[END_ENT]
[CLARIFY: Jordan is a Middle Eastern country...], visited...

Target Output:

Q810

4.3.3 Tokenizer and Model Configuration

The T5 tokenizer is extended with special tokens ([START_ENT], [END_ENT], [CLARIFY:]). Inputs are truncated to 512 tokens.

Model Dimensions

- Hidden size: 768
- Feed-forward layer size: 3072

Transformer Configuration

- Number of encoder/decoder layers: 12
- Number of attention heads: 12
- Maximum relative attention distance: 128

Regularization

- Dropout rate: 0.1

4.3.4 Training Setup

Models are fine-tuned using HuggingFace **Trainer** with AdamW optimizer, learning rate 3×10^{-4} , gradient accumulation, mixed-precision training (FP16), and early stopping based on validation performance.

5 Evaluation Strategy

The evaluation compares *baseline* and *clarified* T5 models. Both models share identical architectures and hyperparameters, where the only difference is LLM-generated clarifications.

5.1 Experimental Setup

Data: 231 test documents, separated from training/validation.

Models:

- Baseline T5 model (without clarifications)
- Clarified T5 model (with LLM-generated clarifications)

5.2 Evaluation Procedure

1. **Model Preparation:** Load trained checkpoints, initialize tokenizers with special tokens, set models to evaluation mode.
2. **Test Data Preparation:** Tokenize inputs and batch (batch size 16).
3. **Prediction Generation:** Generate predictions via beam search and decode QIDs.
4. **Metric Calculation:** Exact Match Accuracy: percentage of exactly correct QID predictions.

5.3 Evaluation Results

The evaluation results, illustrated in Figure 9, show that the baseline model correctly predicted 2,025 out of 4,483 mentions, achieving an accuracy of 45.17%. In comparison, the clarified model correctly predicted 1,858 mentions, with an accuracy of 41.45%. This corresponds to a net decrease in performance, with an absolute change of -3.73% and a relative change of -8.25%.

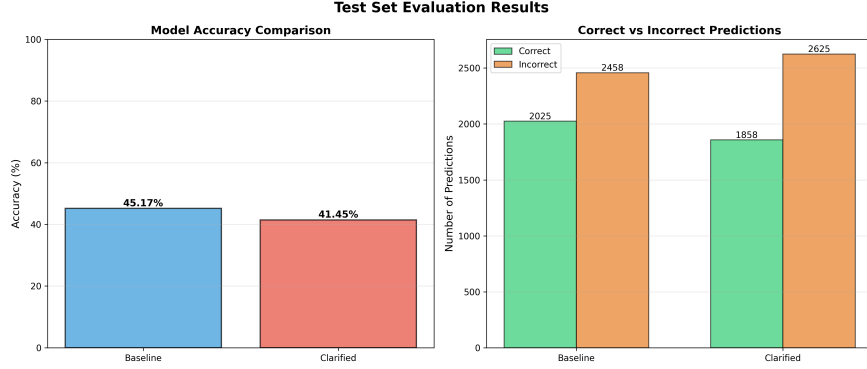


Figure 9: Comparison of Models' Accuracy

5.4 Error Analysis

To understand model behavior, we analyzed prediction errors, focusing on where the baseline and clarified models differ. Predictions were categorized into four groups, summarized in Table 2:

Table 2: Error Analysis Summary

Category	Count (%)
Both correct	1,775 (39.6%)
Both incorrect	2,375 (53.0%)
Only baseline correct	250 (5.6%)
Only clarified correct	83 (1.9%)

While the clarified model improved predictions in 83 cases, it decreased performance in 250 cases, resulting in a net loss of 167 correct predictions. This visualization highlights specific error patterns and identifies opportunities for refining the clarification strategy.

5.4.1 Example Predictions

Clarifications Improving Predictions: The first set of examples, where clarifications helped the model make correct predictions, is shown in Figure 10.

Example 1/5 Input (first 200 chars): link entity: CRICKET - [START_ENT] PAKISTAN [END_ENT] [CLARIFY: **Country**: PAKISTAN is a country 1..... Baseline Prediction: Q843 X Clarified Prediction: Q182538 ✓ Ground Truth: Q182538
Example 2/5 Input (first 200 chars): link entity: 240 4-247 5-252 6-260 7-261 8-269 9-276 Bowling : Doull 8-1-60-0 (w-3) , Kennedy 3-0-..... Baseline Prediction: Q1393201 X Clarified Prediction: Q1766640 ✓ Ground Truth: Q1766640
Example 3/5 Input (first 200 chars): link entity: SOCCER - ENGLISH F.A. CUP SECOND ROUND RESULT . LONDON 1996-12-06 Result of an English Baseline Prediction: Q3344790 X Clarified Prediction: Q48947 ✓ Ground Truth: Q48947
Example 4/5 Input (first 200 chars): link entity: ather in the hope of witnessing one last moment of magic . Campese will be up against a..... Baseline Prediction: Q34647 X Clarified Prediction: Q5465 ✓ Ground Truth: Q5465
Example 5/5 Input (first 200 chars): link entity: 1d reporters . League title-holders Steaua Bucharest , who finished bottom of their Cha..... Baseline Prediction: Q208399 X Clarified Prediction: Q495299 ✓ Ground Truth: Q495299

Figure 10: Examples where LLM-generated clarifications improved predictions.

Cases Where Baseline Was Better: The first set of examples where the baseline outperformed the clarified model is illustrated in Figure 11.

Example 1/3
Input (first 200 chars): link entity: RUGBY UNION - CUTTITTA BACK FOR ITALY AFTER A YEAR . [START_ENT] ROME [END_ENT] 1996-12..... Baseline Prediction: Q220 ✓ Clarified Prediction: Q229 X Ground Truth: Q220
Example 2/3
Input (first 200 chars): link entity: 9-276 Bowling : Doull 8-1-60-0 (w-3) , Kennedy 3-0-24-0 (w-7 nb-1) , Cairns 8-1-35-..... Baseline Prediction: Q982801 ✓ Clarified Prediction: Q3535504 X Ground Truth: Q982801
Example 3/3
Input (first 200 chars): link entity: 9-0-53-1 (w-1 nb-1) New Zealand innings B. Young c Moin Khan b Waqar 5 C. Spearman c Baseline Prediction: Q982801 ✓ Clarified Prediction: Q3535504 X Ground Truth: Q982801

Figure 11: Examples where the baseline model made correct predictions while the clarified model failed.

5.5 Discussion

Although the clarified model succeeded in select cases, overall it performed worse than the baseline. Some LLM-generated clarifications may introduce ambiguity or distract the model. Future strategies could target only ambiguous or difficult mentions to improve disambiguation performance.

Final Summary:

- Baseline Accuracy: 45.17%
- Clarified Accuracy: 41.45%
- Net change: -3.73% absolute

6 Outcomes and Conclusions

The Clarify-and-Link framework demonstrates the potential of using LLM-generated clarifications to improve entity disambiguation. In practice, however, providing clarifications indiscriminately led to a net decrease in overall accuracy. While clarifications can be helpful in highly ambiguous contexts, they may introduce noise or distract the model in simpler cases. A significant factor affecting performance is the design of the prompts used to generate clarifications; if the LLM does not fully understand the instructions, the produced clarifications may be incomplete or inaccurate. Another limitation is the text provided for clarification: context windows that are too small may truncate critical information, preventing the model from generating accurate descriptions of entities.

Analysis of the results indicates that clarifications are most effective for polysemous mentions or cases where additional context is necessary to resolve ambiguity. Conversely, indiscriminate clarification, especially in straightforward mentions, can overwhelm the model and reduce accuracy. Error analysis and example predictions highlight specific patterns and suggest that targeted improvements in clarification

7 References

References

Daniel Vollmers, Hamada Zahera, Diego Moussallem, and Axel-Cyrille Ngonga Ngomo. Contextual augmentation for entity linking using large language models. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 8535–8545, Abu Dhabi, UAE, 2025. Association for Computational Linguistics.